Exploring NYC Yellow Taxi Trips: Unveiling Insights Through Data Analysis

Pradyutha Kaushik · Follow
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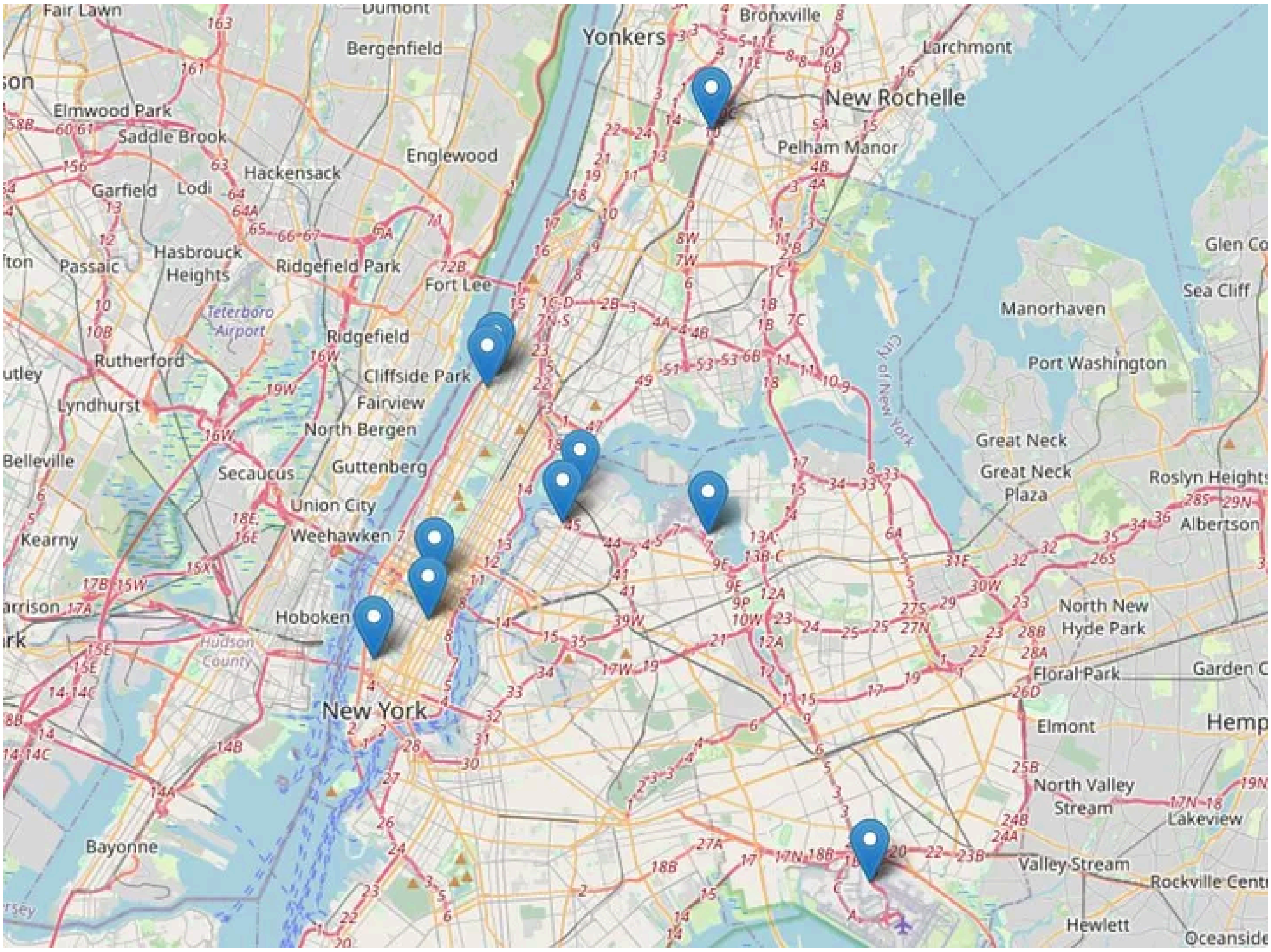


In this article, we delve into a captivating exploration of New York City's Yellow Taxi dataset from the Taxi & Limousine Commission (TLC), aiming to apply advanced analytical techniques to unravel profound insights into usage patterns, spatial dynamics, temporal trends, and regulatory landscapes that govern this bustling metropolis's transportation network.

Our project employs Exploratory Data Analysis, sophisticated clustering, classification, regression, and time-series techniques for a multifaceted analysis to gain meaningful insights.
Yellow Taxis, known for their iconic yellow medallions, operate primarily through spontaneous street hails.

Reviewing Prior Work
Before diving into our analysis, we conducted background research to explore existing studies and projects related to NYC taxi data analysis. Several notable works highlight insights into taxi usage patterns, fare dynamics, and demand forecasting. However, our project aims to bring novel methodologies and perspectives to the table, focusing on clustering, regression, and time-series analyses tailored to the unique characteristics of the dataset.

Primary Objectives
1. Cleaning and Preparing the Data
2. Analysis and ML Models Implementation
3. Time-Series Analysis
4. Providing Actionable Insights and Recommendations

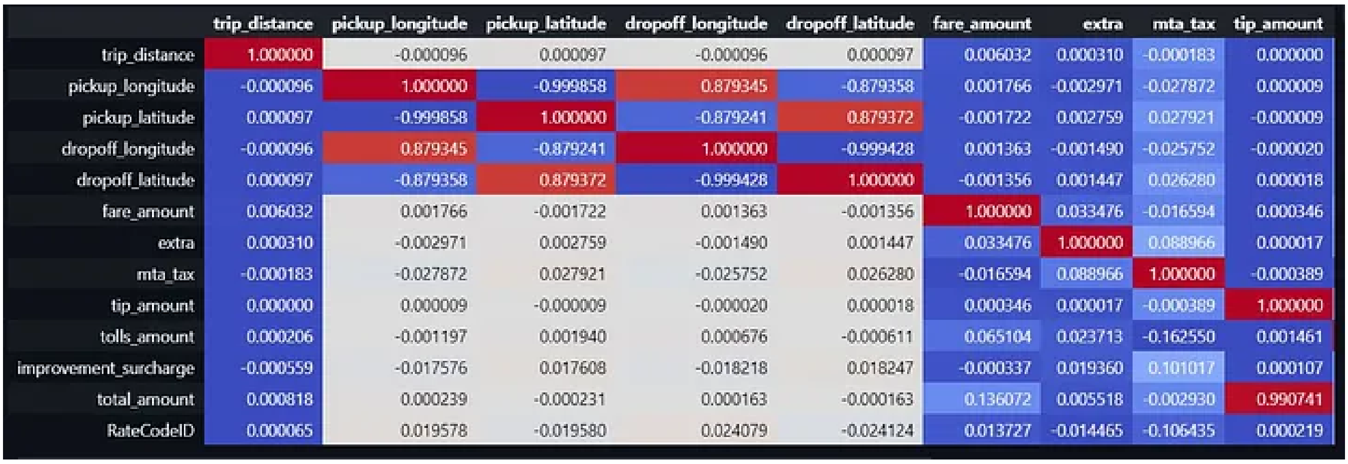
EXPLORATORY DATA ANALYSIS
Exploratory Data Analysis (EDA) is a crucial initial step in any data analysis project. It involves understanding the dataset's structure, identifying patterns, and detecting anomalies before diving into more complex analyses. Let's walk through our EDA process using the NYC Yellow Taxi dataset.

➢ Data Loading and Preprocessing
First, we loaded the Yellow Taxi data for January 2016 to March 2016. Additionally, we imported data for January 2015 to compare temporal trends. For easy manipulation, we consolidated the datasets and converted date-time columns into DateTime objects.
➢ Data Cleaning and Feature Selection
To streamline our analysis, we dropped unnecessary columns and handled missing values.
taxi_df.drop([“tpep_pickup_datetime”, “tpep_dropoff_datetime”,

"store_and_fwd_flag"], axis=1,inplace=True)
taxi_df.drop('Unnamed: 0', axis=1, inplace=True)
# Fill missing values for RateCodeID and improvement_surcharge
taxi_df['RateCodeID'] = taxi_df['RateCodeID'].fillna(1)
taxi_df['improvement_surcharge']=taxi_df['improvement_surcharge'].fillna(0.3)

➢ *Correlation Analysis*

Next, we examined correlations between numerical features to identify potential relationships. The correlation heatmap helps visualize how different features correlate with each other.



	trip_distance	pickup_longitude	pickup_latitude	dropoff_longitude	dropoff_latitude	fare_amount	extra	mta_tax	tip_amount
trip_distance	1.000000	-0.000096	0.000097	-0.000096	0.000097	0.006032	0.000310	-0.000183	0.000000
pickup_longitude	-0.000096	1.000000	-0.999858	0.879345	-0.879358	0.001766	-0.002971	-0.027872	0.000009
pickup_latitude	0.000097	-0.999858	1.000000	-0.879241	0.879372	-0.001722	0.002759	0.027921	-0.000009
dropoff_longitude	-0.000096	0.879345	-0.879241	1.000000	-0.999428	0.001363	-0.001490	-0.025752	-0.000020
dropoff_latitude	0.000097	-0.879358	0.879372	-0.999428	1.000000	-0.001356	0.001447	0.026280	0.000018
fare_amount	0.006032	0.001766	-0.001722	0.001363	-0.001356	1.000000	0.033476	-0.016594	0.000346
extra	0.000310	-0.002971	0.002759	-0.001490	0.001447	0.033476	1.000000	0.088966	0.000017
mta_tax	-0.000183	-0.027872	0.027921	-0.025752	0.026280	-0.016594	0.088966	1.000000	-0.000389
tip_amount	0.000000	0.000009	-0.000009	-0.000020	0.000018	0.000346	0.000017	-0.000389	1.000000
tolls_amount	0.000206	-0.001197	0.001940	0.000676	-0.000611	0.065104	0.023713	-0.162550	0.001461
improvement_surcharge	-0.000559	-0.017576	0.017608	-0.018218	0.018247	-0.000337	0.019360	0.101017	0.000107
total_amount	0.000818	0.000239	-0.000231	0.000163	-0.000163	0.136072	0.005518	-0.002930	0.990741
RateCodeID	0.000065	0.019578	-0.019580	0.024079	-0.024124	0.013727	-0.014465	-0.106435	0.000219

Exploring the dataset through EDA sets the stage for deeper analyses using machine learning and statistical techniques.

FEATURE SELECTION AND REGRESSION ANALYSIS

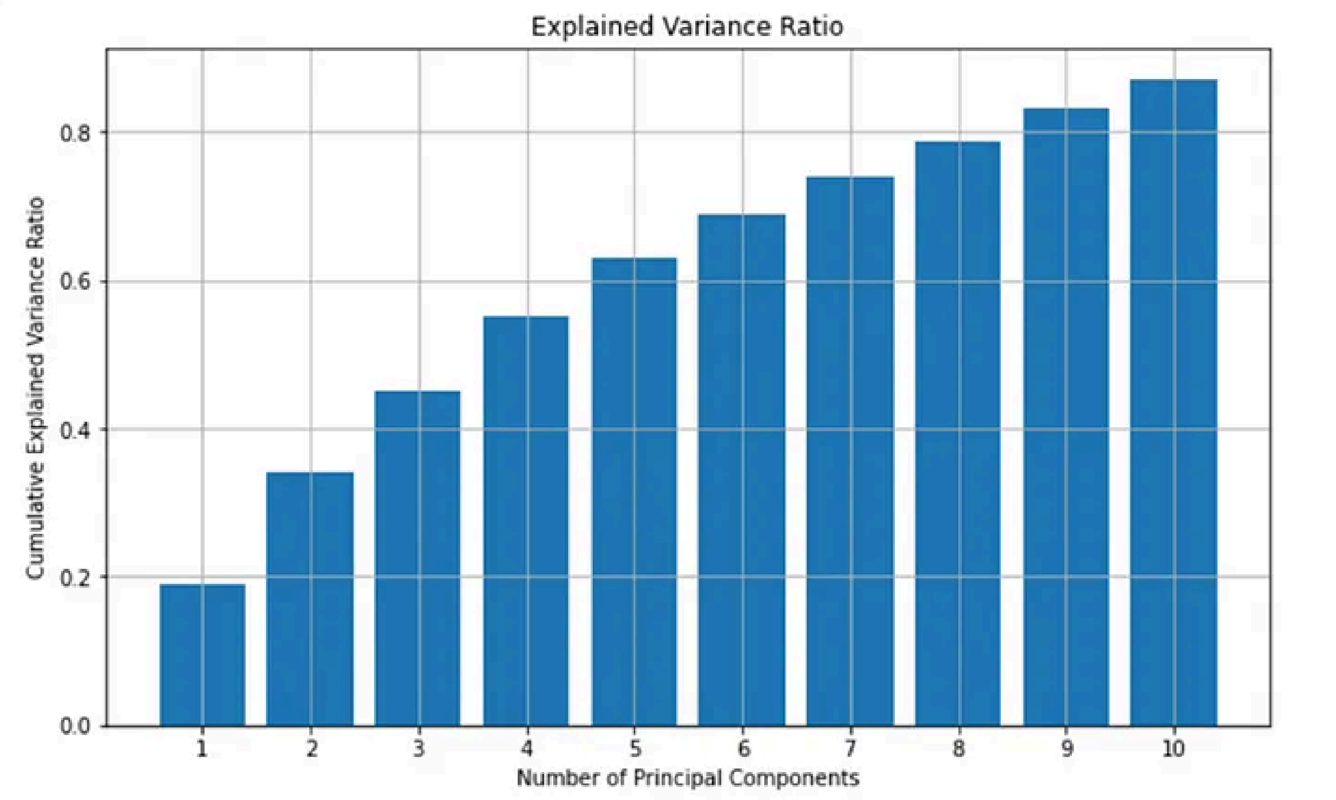
In this analysis section, we dive deeper into feature selection and regression techniques to optimize fare predictions for New York City (NYC) taxi trips. Leveraging advanced machine learning models, we aim to predict fare amounts accurately based on relevant trip attributes.

➢ *Feature Selection and Data Preprocessing*

We start by loading a sample of the consolidated taxi dataset and preparing it for analysis. Unnecessary datetime columns (pickup_time and dropoff_time) are dropped for simplicity. Categorical variables like VendorID, payment_type, and RateCodeID are encoded into dummy variables using one-hot encoding.

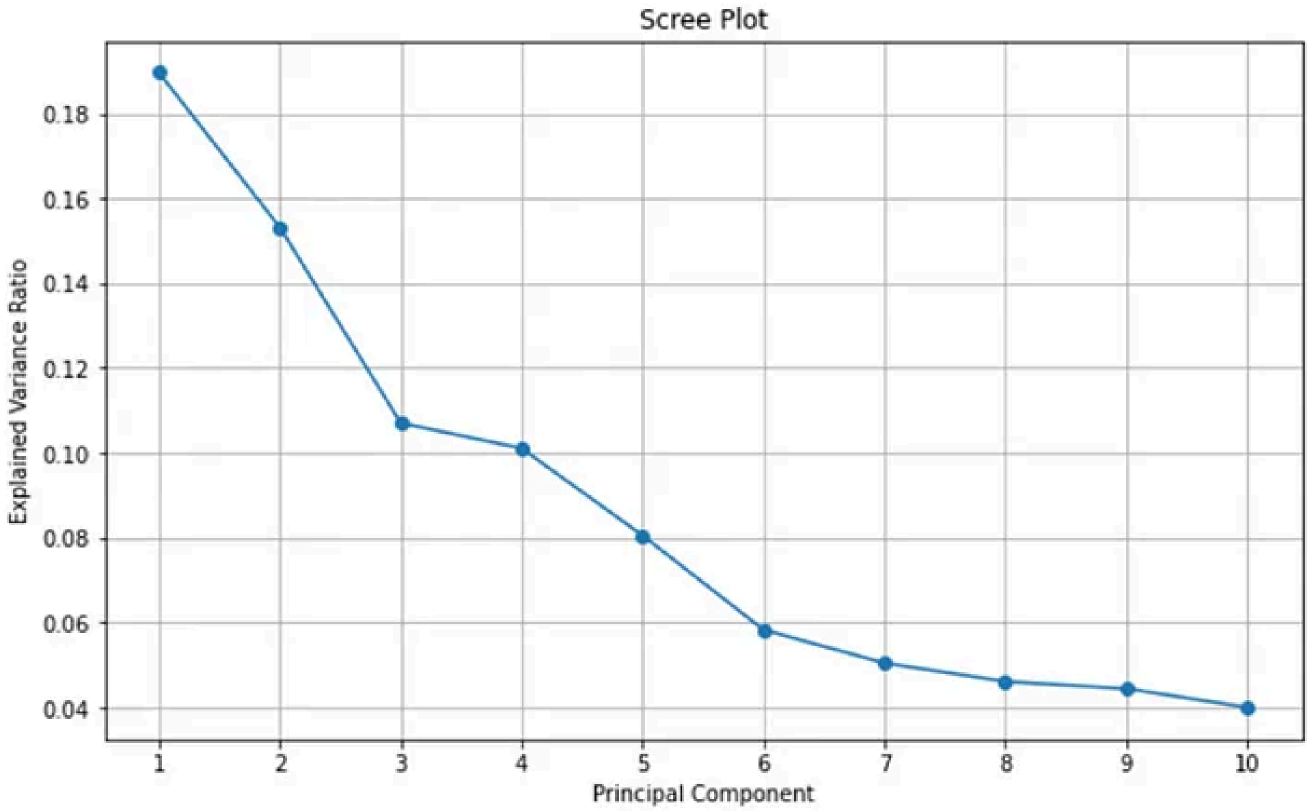
➢ *Standardization and Principal Component Analysis (PCA)*

PCA is beneficial for reducing the dimensionality of the dataset while retaining the most essential information. High-dimensional datasets like NYC taxi trip data can have many interrelated features. PCA allows us to capture the variance in the data using a smaller number of principal components. This dimensionality reduction simplifies the modeling process, improves computational efficiency, and helps avoid issues like overfitting that can arise with many features.



Variance Plot:

- CEVR indicates the proportion of variance in the dataset explained by each principal component. The plot shows that the first few principal components explain significant variance, gradually decreasing as we move to subsequent components.
- For example, here we observe a maximum CEVR of 0.8 for the first 10 features, which implies that these ten principal components collectively explain 80% of the total variance in the dataset.
- The variance plot helps decide how many principal components to retain based on the desired level of explained variance. In your case, retaining a certain number of principal components (e.g., the first 10) might be sufficient to capture a substantial amount of variance while reducing dimensionality.



1. Scree Plot:

- In this plot, EVR indicates the proportion of variance explained by each principal component. A higher EVR value for a principal component signifies that it captures more information from the dataset.
- In this scree plot, PC3 and PC4 have similar EVR values around 0.10 and 0.11, respectively, suggesting that they contribute equally to explaining the variance.
- The scree plot is instrumental in identifying the "elbow point," where the rate of decrease in variance explained by additional principal components becomes less significant. This helps in deciding how many principal components are necessary to retain. In our case, to accommodate more diversity, we choose ten as our count.

We perform standardization and dimensionality reduction using PCA to prepare our features (X) for regression, and we apply Principal Component Analysis (PCA) to reduce the dimensionality of the dataset while retaining important information. The selected_feature_names list contains the names of the top principal components selected by PCA.

➢ ***Regression Model Comparison & Evaluation***

In analyzing New York City (NYC) taxi trip data, regression is a fundamental tool for predicting fare amounts based on various trip attributes.
Some aspects of why regression is essential in the analysis are for the purposes:
• Fare Estimation
• Understanding Factors Influencing Fares
• Modeling Complex Relationships

Regression models can capture these nonlinear relationships and provide a systematic way to quantify their impact on fare amounts.

```
Linear Regression: Mean RMSE = 4.847812532295883, Standard Deviation = 0.596805804367288
Lasso Regression: Mean RMSE = 4.858269345080977, Standard Deviation = 0.5826021425996598
Ridge Regression: Mean RMSE = 4.847812539417645, Standard Deviation = 0.596805656868091
```

We compare the performance of different regression models (Linear Regression, Lasso Regression, and Ridge Regression) based on Root Mean Square Error (RMSE) metrics.
Ridge Regression was selected as the best model for predicting taxi fare amounts due to its effectiveness in handling multicollinearity, balancing bias and variance, and providing robust performance on test data. Multicollinearity can lead to unstable and unreliable estimates in traditional regression models. The regularization technique employed by Ridge Regression helps stabilize the model by penalizing significant coefficients, which is particularly beneficial for datasets with correlated predictor variables like those found in NYC taxi trip data.

```
Test RMSE of the best model: 6.244492656026829
```

By leveraging Ridge Regression, we can derive meaningful insights into fare estimation and contribute to enhancing the efficiency and effectiveness of taxi services in NYC.

CLASSIFICATION
Classification plays a multifaceted role in the context of Yellow Taxi transportation, such as the categorization of taxi trips based on specific attributes such as pickup time, day of the week, and time of day, which would, in turn, through model training, allow for the prediction of the category or class to which a new taxi trip belongs based on its features. Classification outputs could also provide actionable

information for optimizing taxi services, such as predicting peak demand periods (e.g., morning rush hour, weekend evenings) and allocating resources accordingly. The classification work was hinged on developing a model to enhance decision-making within the taxi industry by predicting the potential revenue classification based on operating within a net of latitude/longitude combinations linked to a given geographical sphere at a specific time and day of the week.

Previous work on Yellow Taxi research has entailed model framework development based on integrated linear regression (LR) long-short-term memory(LSTM), which sought to forecast the demand for taxi rides.

(Kim et al., 2020). The current work, however, hopes to supplement previous work by addressing gaps by introducing classification while upholding simplistic and much more interpretable models. This work was initiated by feature selection which sought to avail the most impactful independent variables, followed by identifying a net of potential model types inclusive of XGBoost, Linear Discriminant Analysis, and ADA Boost. Before model selection, new features were created to aid the modeling exercise, and this was coupled with dummy coding, among other techniques.

➢Algorithm Selection

Algorithm selection was conducted with accuracy, simplicity, and efficiency in mind. Of the three algorithms (XG Boost, Linear Discriminant Analysis, ADA Boost), the XG Boost algorithm was selected based on its capabilities to handle missing values which is inevitable with the nature and size of the dataset. Additionally, XG Boost's combination of accuracy and speed amongst its counterpart boosting algorithms was considered valuable in this work. Linear Discriminant Analysis has inbuilt

dimensionality reduction capabilities, which were thought to suppress dataset complexity and should further work include vast dimensionality. Such a move was envisaged to enhance computational efficiency.

The ADA boost model was selected based on a combination of high accuracy and limited hyperparameters to tune. In most cases, the defaults yield desirable results, thus speaking to simplicity. Furthermore, ADA Boost handles class imbalances by uplifting the importance

of minority classes. These aforementioned 3 Algorithms were thus employed in model development.

➢Feature Creation and Selection

As part of the model development exercise, features for the task at hand were selected based on firstly identifying the target variable, which in this case was "encoded_total_amount" which was the encoded version of the

"total_amount_label" (very low, low, medium, and high) which represented the magnitude
of generated revenue per trip. The "total_amount_label" was derived by categorizing the "total_amount" variable into four categories using quartile thresholds like this:

Quartile threshold	total_amount_label
> = 0.75	high
0.75 > total_amount >= 0.5	medium
0.5 > total_amount >= 0.25	low
total_amount < 0.25	very low

This total_amount represents the overall revenue received by a taxi driver in a given trip, and having this as a target was seen to both play the role of having the taxi industry itself predict the expected revenue range for a driver plying at a given time of day, day of week and at a given geographical zone within New York City and also have taxi insurance providers obtain insights on the development of taxi revenue-based insurance policies to allow for equitable policy premium payments.

For example, having a flat fee for all insurance policies would be seemingly unfair for drivers operating in low-income geographical areas and at low-income times of the day and days of the week compared to those operating within a high revenue-generating geographical area and at peak hours. Features selected included the following, 'VendorID', 'passenger_count', 'trip_distance', 'pickup_longitude', 'pickup_latitude', 'RatecodeID', 'dropoff_longitude', 'dropoff_latitude', 'payment_type' and the 'pickup_time' variable. Based on the "pickup_time" variable, four new variables were created, including the (1) "day_of_week_num" which is a numeric version of the day of the week, Monday being denoted as "0" and Sunday being denoted as "6" (2) "day_of_week" which is the name representation of the day of the week, (3) "time" which represents the clock time in 24 hour notation and (4) "time_category" which represents the time of day (morning, afternoon, evening, night).
The "time_category" and "day_of_week" variables were then dummy-coded to facilitate conversion into numeric form for modeling purposes. In total, the features employed for the classification modeling exercise, including the dummy coded variables, were 20.
➢***Model setup and cross-validation***

Three Classification models were developed based on the three selected algorithms. Model training was done using 80% of the dataset and was accompanied by model validation through K-fold cross-validation. With three folds selected, an average was obtained while observing the accuracy scores of the individual folds conducted to identify the propensity of overfitting. Simplistic hyper parameters (defaults) for all models were applied.

➢*Summary of results*

Still, to combine simplicity, accuracy, and efficiency, the ultimate selection of the best model had to be done based on performance metrics inclined toward accuracy. The table is a summary of the classification model performance.

	XG Boost	LDA	ADA Boost
Training Accuracy	0.803163	0.363168	0.750565
Testing Accuracy	0.795644	0.364733	0.748545
Training f_1 score	0.810000	0.340000	0.750000
Testing f_1 score	0.800000	0.340000	0.750000
Cross-validation accuracy	0.795298	0.365321	0.748817

➢*Cross validation*

Training and Testing Accuracy: The training and testing accuracies of the XG Boosting model were 0.803 and 0.796, respectively. These metrics measure the proportion of correctly predicted trip categories.

F1 Score: The F1 score (harmonic mean of precision and recall) for training and testing sets reflects the model's performance balancing precision and recall for trip category prediction.

High Accuracy: The high testing accuracy (0.796) and cross-validation accuracy (0.795) of the XG

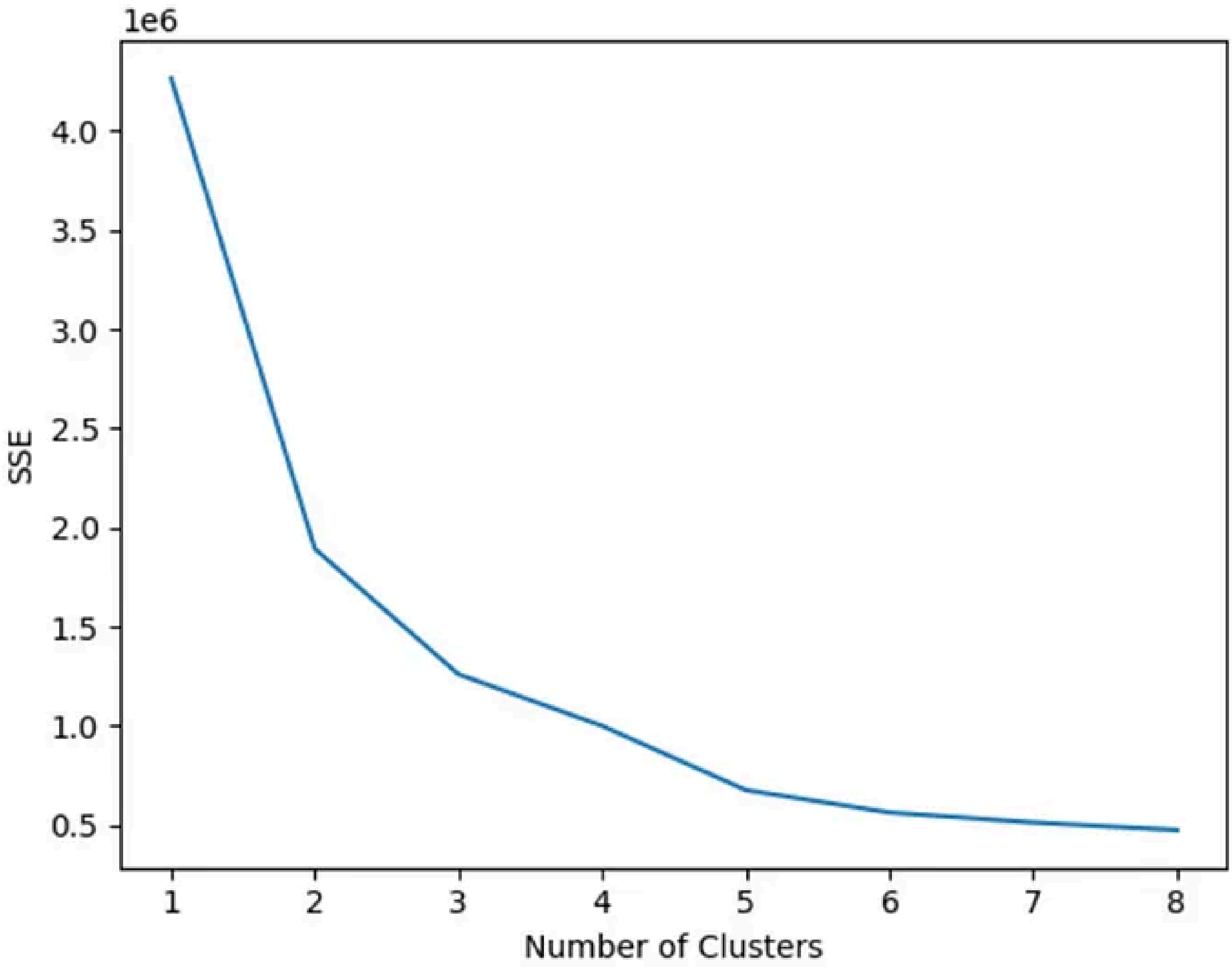
The boosting model indicates its capability to accurately classify taxi trips into meaningful categories based on time-related features.

Classification outputs provide operational insights for taxi companies, enabling them to optimize service delivery, resource allocation, and customer experience based on predicted trip patterns and characteristics.

CLUSTERING

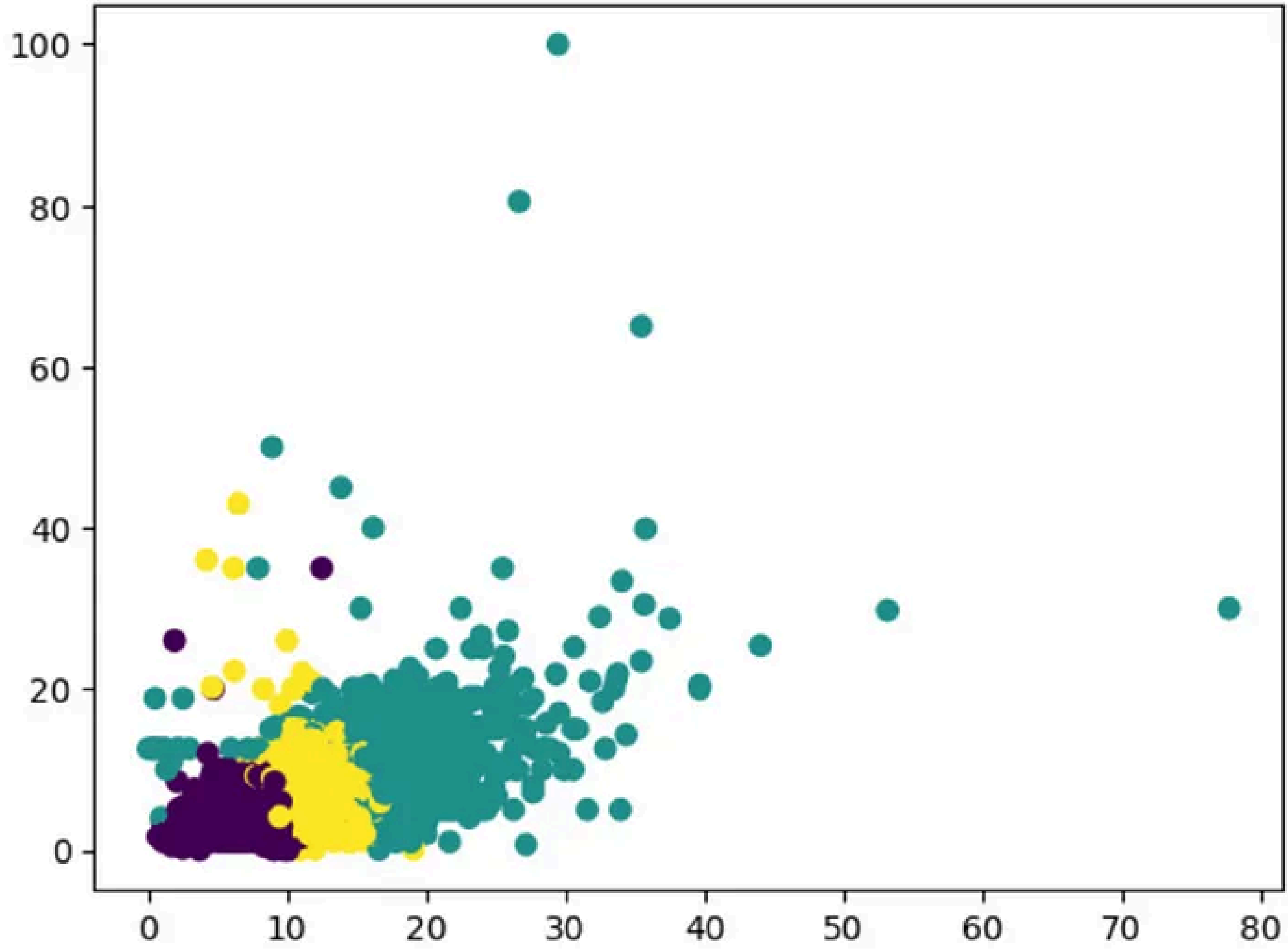
Let's delve into the insights derived from clustering without displaying the specific code to identify inherent patterns and segment data into meaningful groups.

➢ *Determining Optimal Cluster Number*



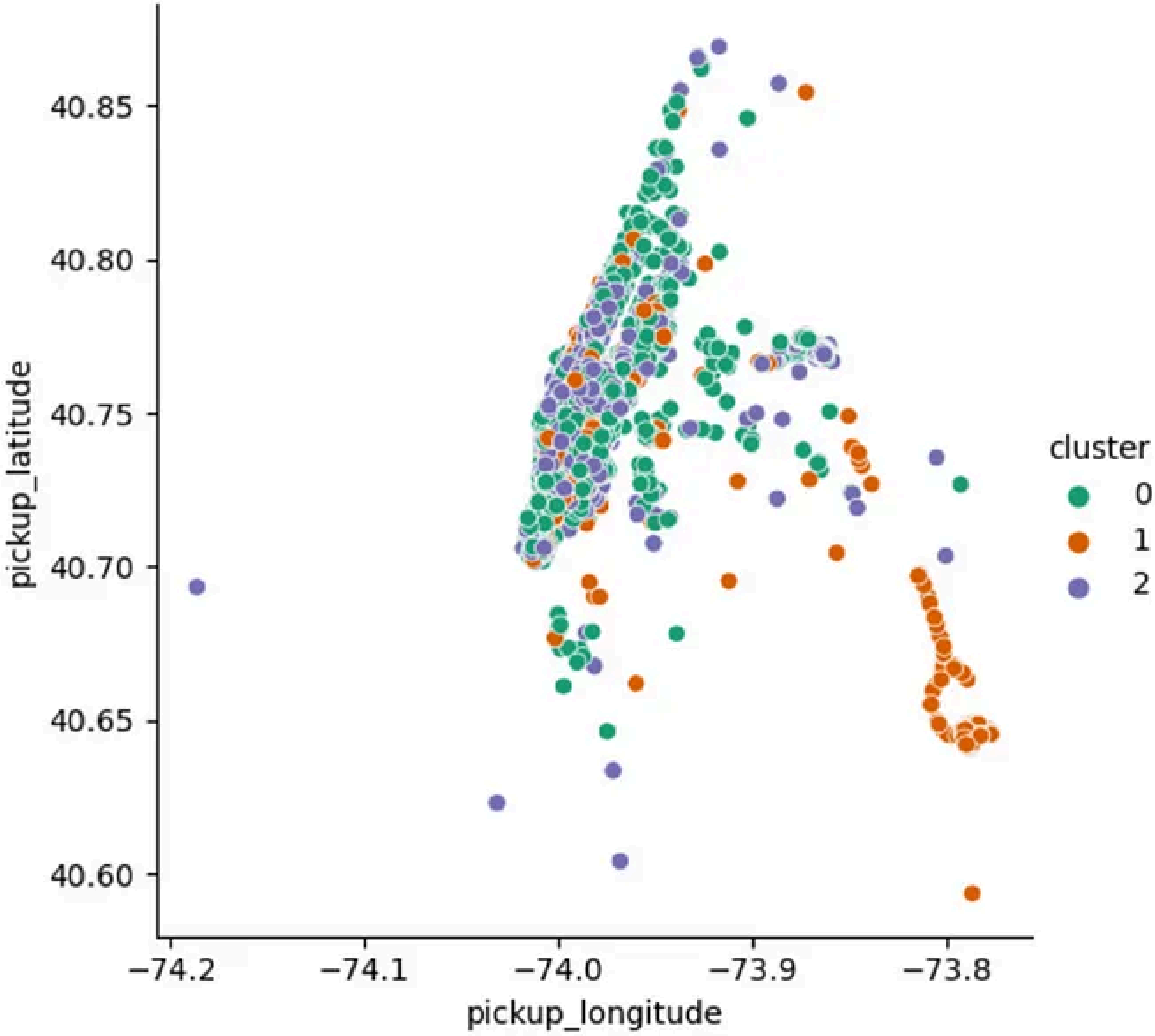
We employ the elbow method to determine our dataset's optimal number of clusters. By plotting the sum of squared errors (SSE) against different cluster numbers, we identify the elbow point that signifies the optimal balance between intra-cluster cohesion and inter-cluster separation.
In this case, we select 3 clusters based on the elbow method analysis.

➢ *Visualizing Clusters in Scatter Plot*



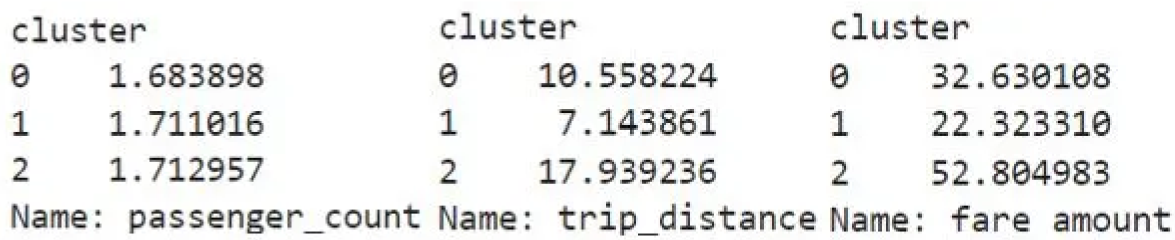
This scatter plot visualizes the clustering results based on trip distance and tip amount. Each point represents a taxi trip, color-coded by its assigned cluster. This visualization provides insights into how trips are distributed across different clusters based on distance and tip amount.

➢ ***Geographic Representation of Clusters***



Using geographic coordinates (longitude and latitude), we create scatter plots that map pickup locations colored by their respective clusters. These plots visualize spatial patterns, revealing clusters of pickup points across New York City.

➢ *Analyzing Cluster Characteristics*



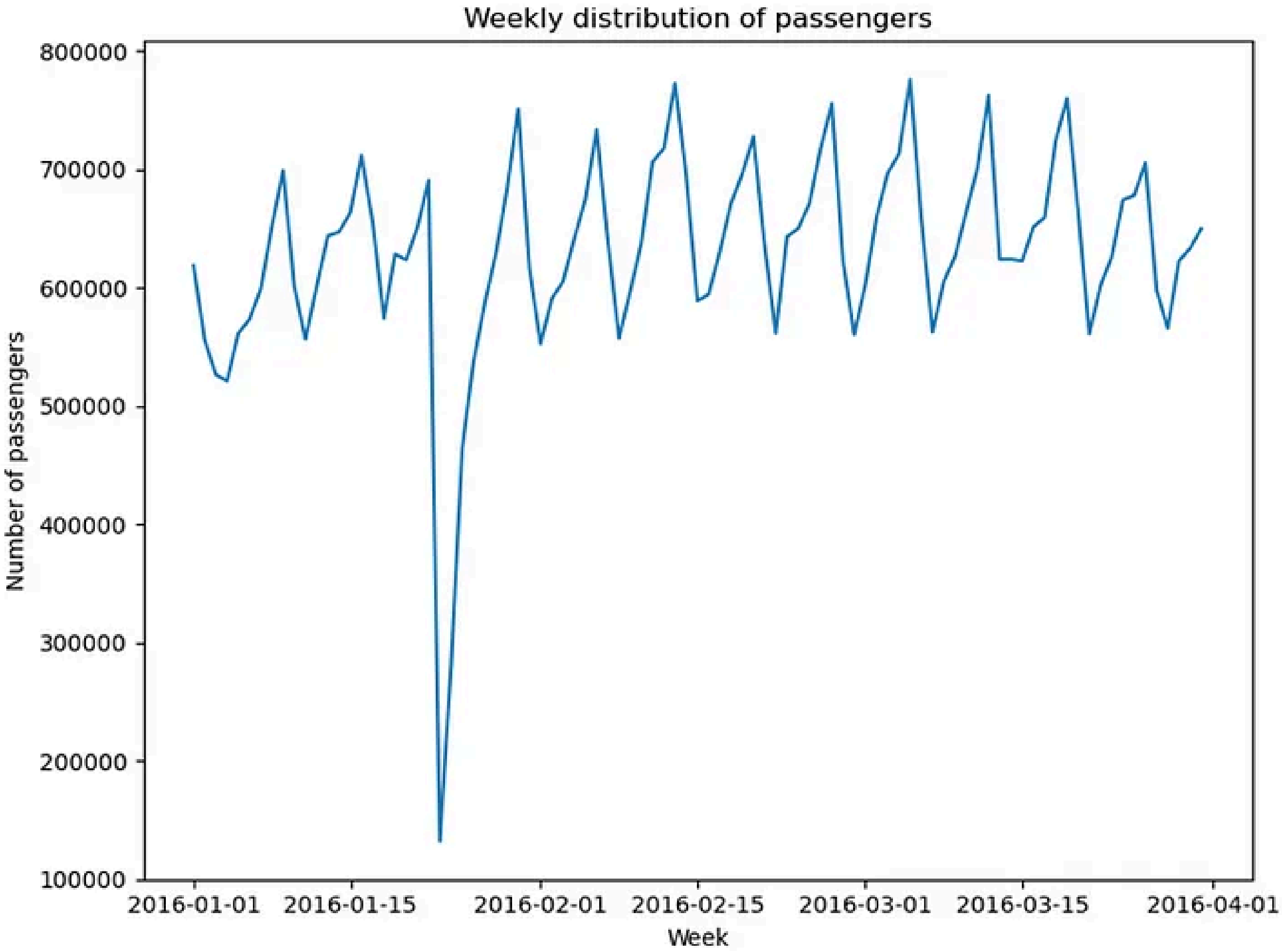
```
cluster                   cluster                 cluster
0    1.683898             0    10.558224          0    32.630108
1    1.711016             1     7.143861          1    22.323310
2    1.712957             2    17.939236          2    52.804983
Name: passenger_count Name: trip_distance Name: fare amount
```

We compute and analyze key metrics such as average trip distance and average number of passengers per cluster. This analysis helps understand each cluster's distinguishing characteristics and behaviors, aiding in targeted interventions or service optimizations.

The insights from clustering analysis contribute to data-driven decision-making in the transportation sector, guiding resource allocation, service design, and operational efficiencies within NYC's dynamic taxi industry.
By leveraging clustering techniques, we uncover hidden patterns and structure within taxi trip data, paving the way for strategic interventions and optimizations in urban transportation services.

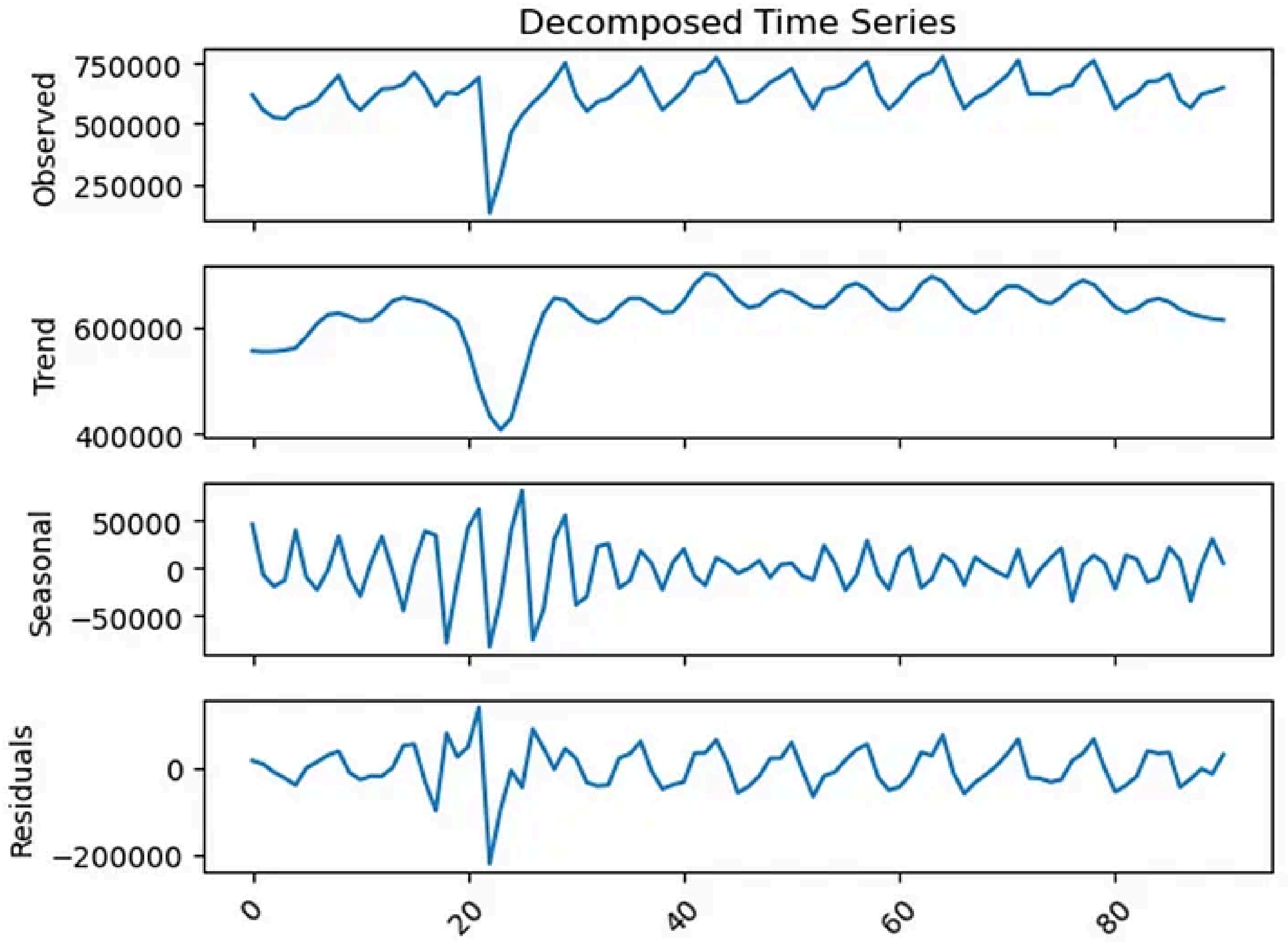
TIME SERIES ANALYSIS
Time series analysis was conducted to evaluate the time dependence of the passenger count, which sought to unearth possible cyclical patterns and thus provide a framework to analyze the New York City taxi ride data that could affect the revenue generated within the industry. The prime benefits gleaned from such an exercise entail imparting knowledge on the evolution of passenger counts over time and demand forecasting. The time series work constituted extracting the pickup_time variable, and passenger counts from the primary dataset followed by applying time series analysis. Previous work conducted on the time series analysis, to a larger extent, focused on the use of spatiotemporal autoregressive (STAR) models (Faghih, 2019; Safikhani *et al*. 2017). however, with the current work, limiting the scope to time series forecasting was prime.

➢Data preparation and visualization
For the time series analysis work, the data employed ranged from January 2016 to March 2016. It should be noted that the recorded passenger counts given were obtained per individual pickup. As such, the passenger count was aggregated into much larger daily chunks using the group by function to facilitate the smoothing out of variability attributed to noise and randomness brought up by peak hours, unusual events and weather conditions. Such an exercise was thought to enhance the identification of
underlying behaviors of the data.



Plot of Number of passengers vs time in weeks

The above image shows the oscillatory nature of the passenger count within the three-month period. It is noteworthy, however, that during the period from 01–15–2016 to 02–01–2016, there was a sharp dip in the passenger count, which was indicative of the underlying pattern.

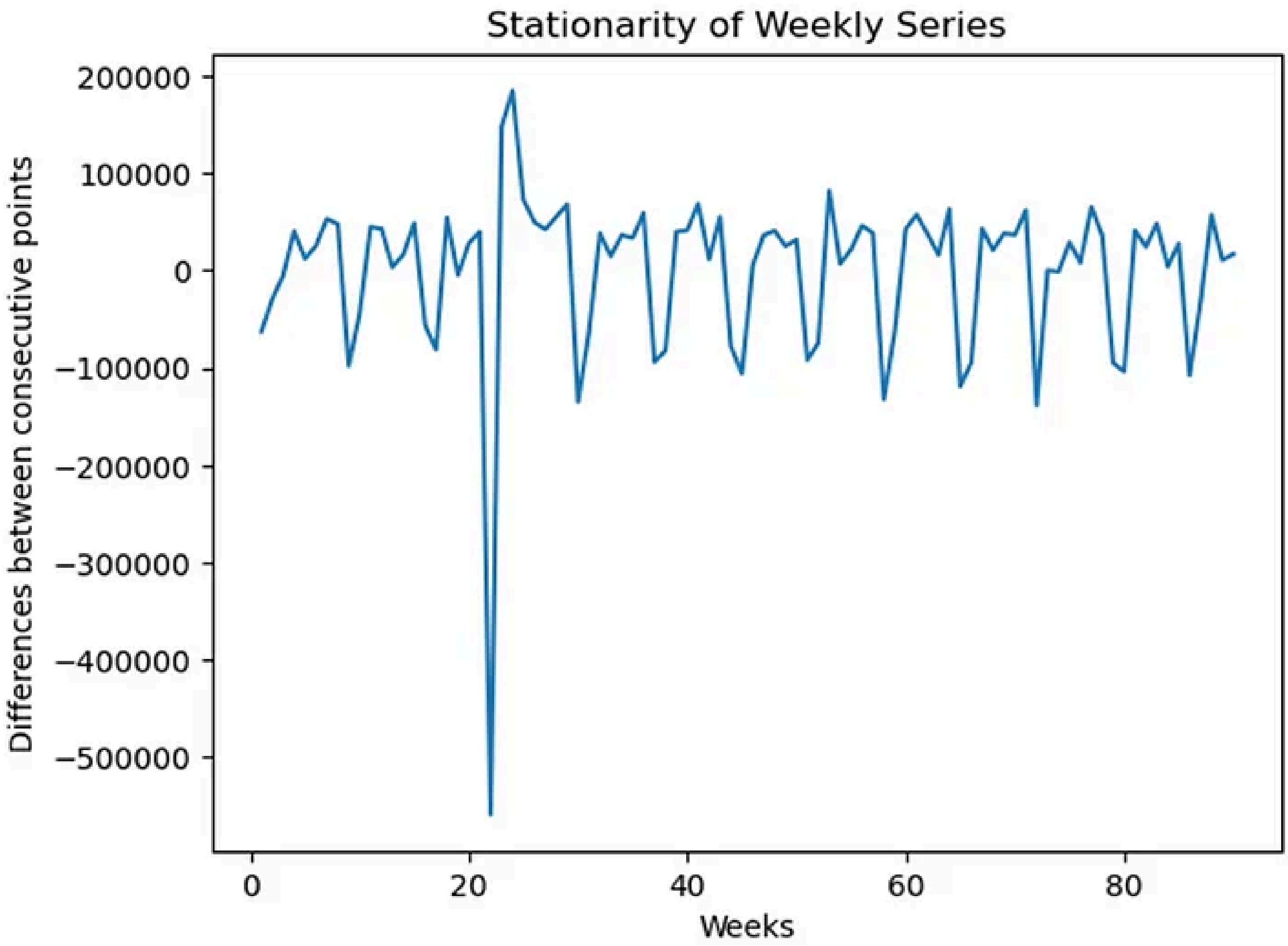


Decomposed time series

The above visualization shows a Decomposed time series for the data, which comprises the observed, the trend, seasonality, and the respective residual plots.

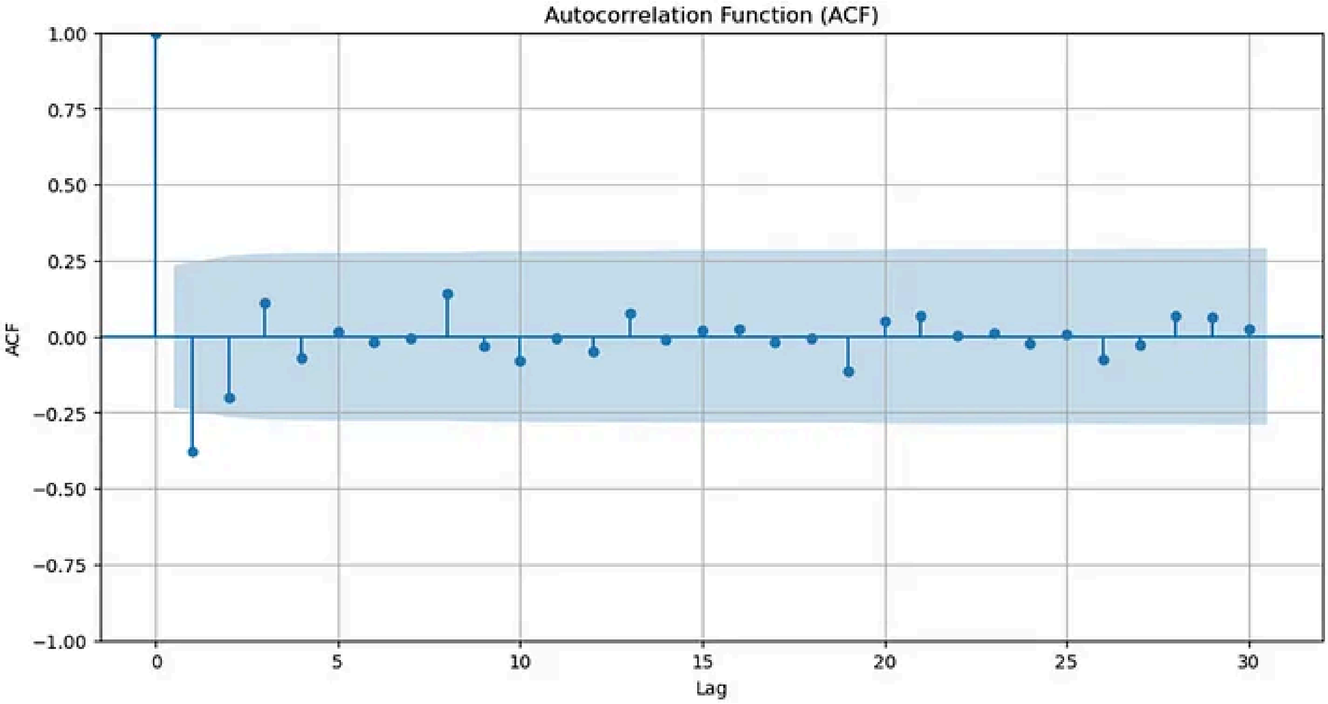
Observing each plot individually, the trend does show a sinusoidal oscillation about a fixed axis, except for the apparent dip observed in the second half of January 2016. In general, there is no evident compound increase in the number of taxi riders on a month-on-month basis, indicating a steady state (no increase or decrease in demand for service usage).
The seasonal plot, through its cyclical nature, does show recurring patterns in the count of taxi riders. This realization gives impetus towards establishing a framework within the taxi industry that allows for effective resource allocation through decongesting the platform during low peak periods while ramping up the number of taxi drivers during weekends. The residual plot presents visualizations of irregularity and noise attributed to unpredictable events.
Through differencing, it was possible to ensure that the series was stationary, thus paving the way for the application of Autoregressive Integrated Moving Average (ARIMA), as this is a requirement for effective

and reliable forecasting. The above image accompanied by results obtained from the Adfuller test, did confirm stationarity.

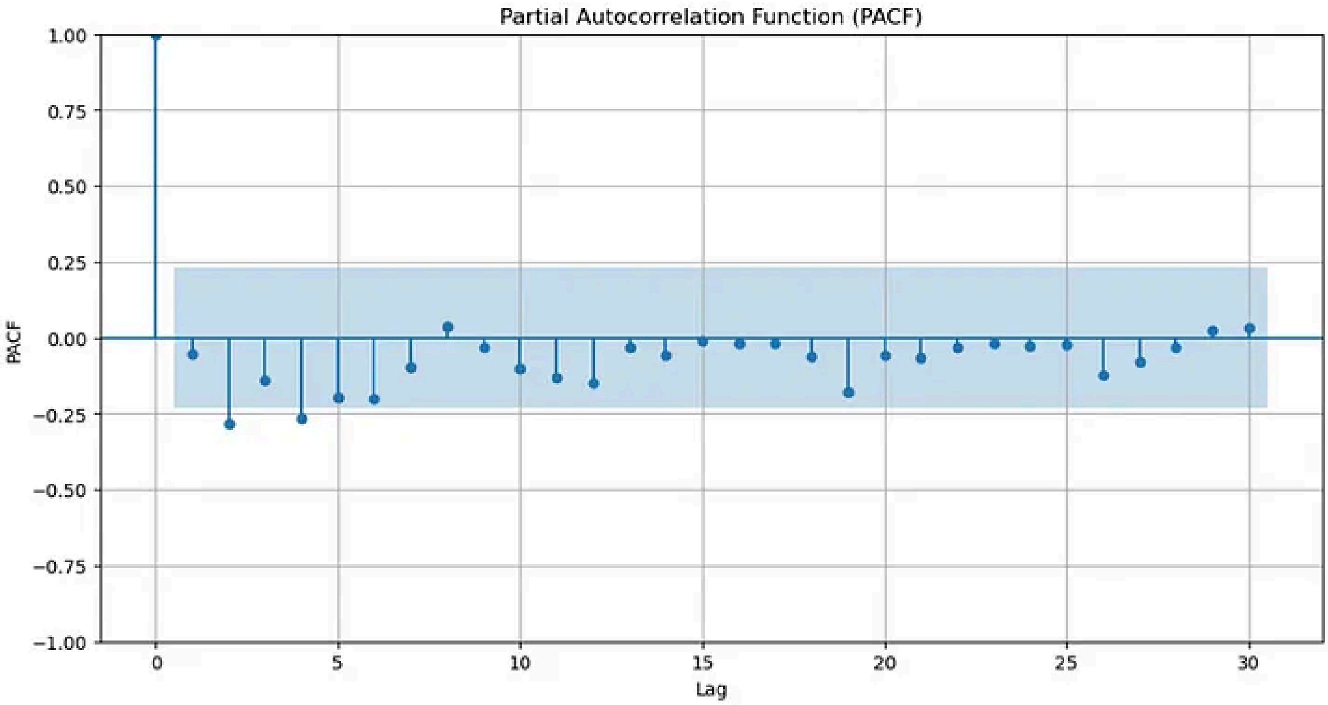


The Adfuller test was applied with the visual stationarity depiction to double-check stationarity. Results obtained from the test yielded an ADF Statistic of -6.86 and a p value of 1.57 x 10-9, satisfying conditions for stationarity (negative ADF statistic and low p-value).

Autocorrelation function

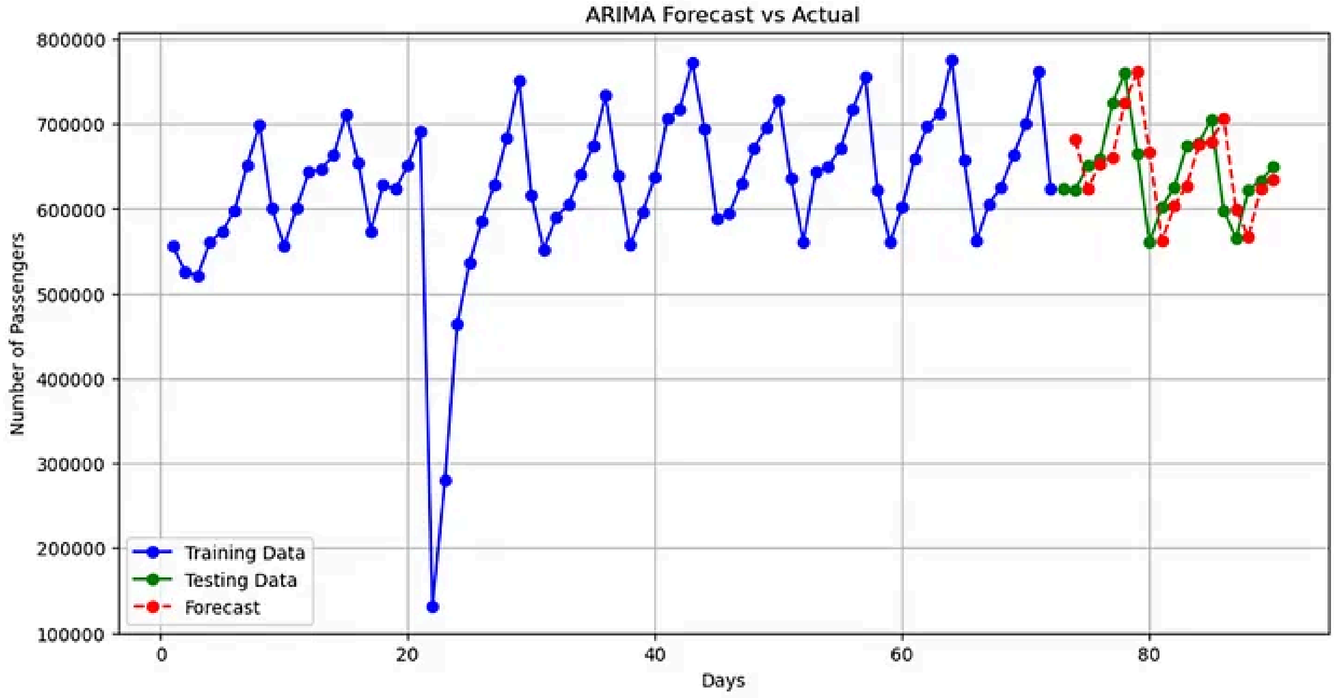


Partial Autocorrelation function

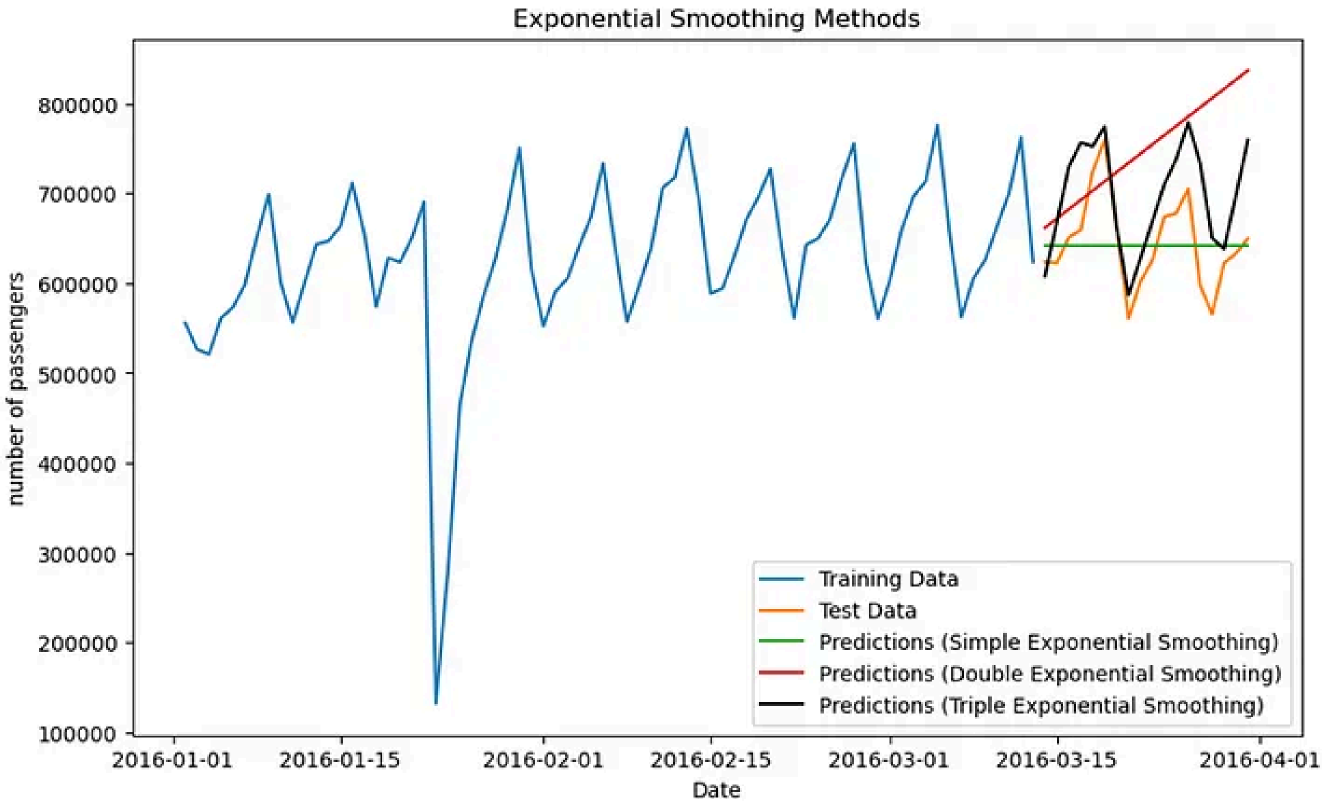


➢*Time series forecasting*

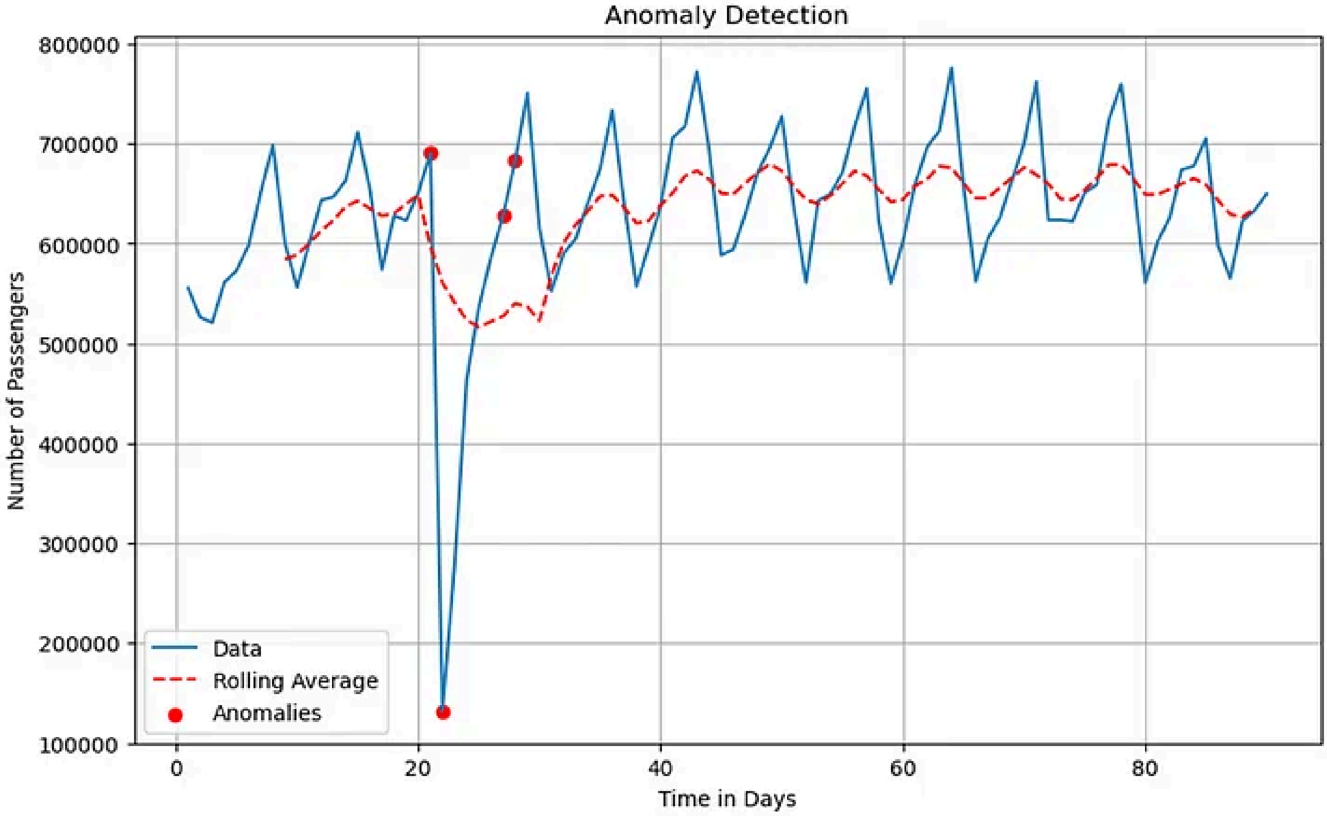
Two forecasting models were selected, these being ARIMA and Exponential smoothing. The selection of these models was hinged on the need for ease of interpretability, ease of use, and robustness to missing values characteristic of the ARIMA model. Additionally, the simplicity offered by exponential smoothing is a property that would ease understanding for novices.



Based on the image, the developed ARIMA model closely fits the data, with forecast values (in red) closely aligning with the actual test values (green). Such a development is envisaged to facilitate close to accurate forecasts of the utilization trajectory of the yellow taxi service, which in turn can be used to implement dynamic pricing during peak/non-peak periods to maximize revenue generation. Additionally, with the knowledge of expected demand gleaned from the ARIMA visual, customer satisfaction can be upheld through the pro-active addition of resources before peak times, thus stemming off wait times through increasing availability combined with the optimization of routes to minimize travel times as will be discussed in network analysis. Lastly, strategic business decisions inclusive of possible expansions and investment priorities are thought to, in part, be hinged on insights drawn from ARIMA models developed from a much broader time; this is a limitation of the currently developed model.



The image depicts Exponential Smoothing, an additional model developed to aid in the forecasting work. Of the models shown, the triple exponential smoothing model closely followed the trend presented by the test data. Such an observation renders the model applicable for predictive efforts. However, comparing model metrics of the Tripple exponential smoothing and Arima models indicates that the ARIMA model far outdoes the exponential smoothing model based on Root Mean Square Errors.



NETWORK GRAPH ANALYSIS

Network Analysis was conducted to facilitate the identification of the most important hubs within the taxi network. Starting off with removing latitude values that were equal to zero, this eliminated nodes that would not add value to the analysis. New columns with tuples of the pick_up and drop_off points were developed using the pickup_latitude/pickup_longitude and dropoff_latitude/dropoff_longitude combinations. Additional columns represented a numerical identity of all nodes within the dataset. Combining all node/hub identities into a single data frame, these were ranked using the page rank algorithm to ascertain their level of importance within the network. Ultimately, plots of network graphs of selected source and target combinations were developed to aid visualization of the network structure.

➢Data Preparation

As part of the exploratory data analysis, pickup latitude/longitude and drop off latitude/longitude columns with latitude/longitude values that were equal to zero were dropped as Global Positioning System (GPS) coordinates were critical in identifying address information. Latitude/longitude combinations for the pickup and drop-off coordinates were each combined into single columns as tuples, yielding columns named “pickup_point” and “dropoff_point”. To pave the way for the application of page ranking, numerical point IDs were assigned to each pickup or drop-off point within the dataset.

The significant figures selected for GPS coordinates were three decimal places corresponding to an accuracy radius of 110 m. This was deemed an acceptable compromise, as one decimal place corresponds to 11.1 km, and four decimal places correspond to 11m.

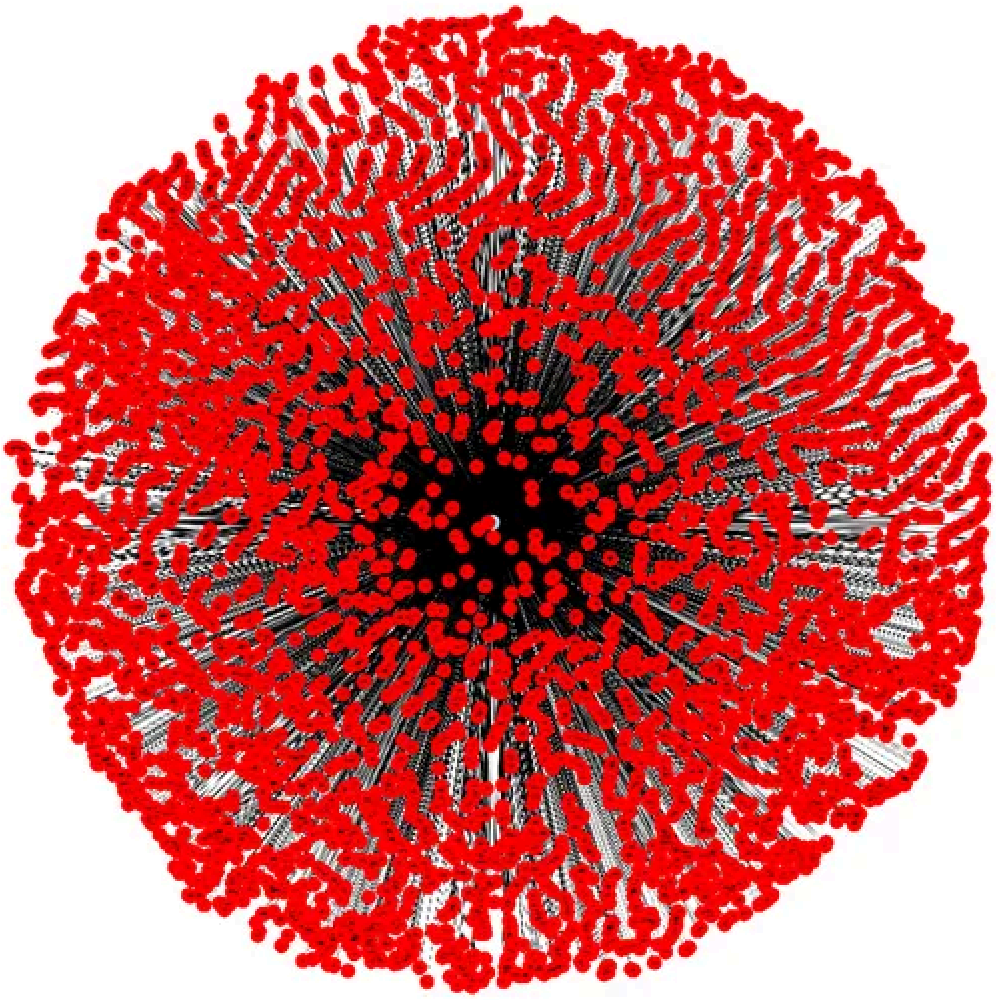
➢Page Ranking

With preparatory work completed, the page rank algorithm was applied to both the “pickup point” and “dropoff point” IDs, which, accompanied by sorting, resulted in a data frame that had the highest ranked nodes or most influential taxi pick and or drop-off spots within New York City. Concatenation of corresponding GPS coordinates for the highest ranked nodes was followed by scrapping the https://nominatim.openstreetmap.org webpage for address information that matches the point coordinates shows the top ten most influential points within the New York City Yellow Taxi network.

point_ids	page_rank	location_address
68	0.01014	John F. Kennedy International Airport
49	0.00855	John F. Kennedy International Airport
12	0.00430	LaGuardia Airport, Marine Terminal Road
411	0.00365	LaGuardia Airport, Marine Terminal Road
11	0.00325	Terminal 8 Departures
44	0.00265	Terminal C, Terminal C Outer Arrivals
537	0.00256	JFK Access Road
61	0.00220	Terminal C, Terminal C Outer Arrivals

It is indicative from the above image that the most critical nodes circle the John F Kennedy and LaGuardia airports. Such an observation is believed to guide initial work toward Yellow Taxi route optimization. Without such knowledge, route optimization could be inadvertently implemented in areas where the need is not justifiable.

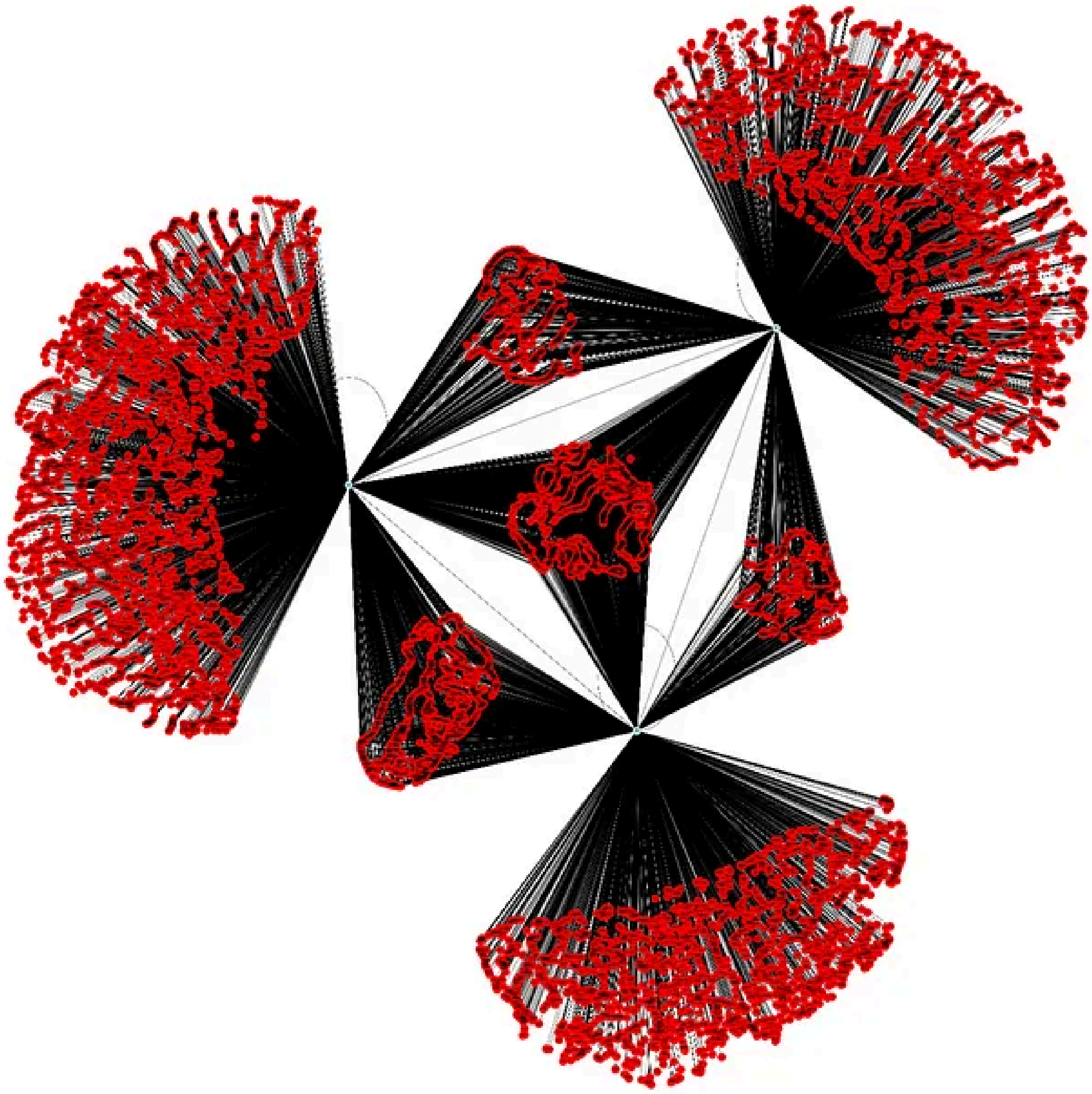
➢Network Graphs



Network graph of the highest-ranked node corresponding to John F Kennedy Airport (JFK).

The image shows that the JFK airport has a high level of centrality and the density of connected nodes is indictive of high connectivity. Such information is thought to guide initial work toward optimizing the

Yellow taxi network through a myriad of strategies inclusive of the expansion of infrastructure through availing alternative exits routes out of the airport, such as additional electric trains that would transfer passengers to multiple distribution points where they could have their taxi rides to their final destinations. The latter is thought to create new hubs in the network, thus decentralizing passenger pick-up and/or delivery from the airports.



Network structure around the three highest-ranked nodes

This is the network graph of the three highest-ranked nodes. These correspond to two locations within the John F Kennedy airport, and the third point is LaGuardia Airport.

Quite interestingly, there are miniature cliques of nodes within the center of the graph. The density of clusters around each of the three hub nodes is comparable, suggesting that any decongestion work could be applied equitably, that is, the allocation of resources for such an exercise.

REFLECTION

Our journey through analyzing NYC taxi trip data using clustering techniques has

been enlightening, providing valuable learnings and insights into urban transportation dynamics. Some reflections on our
key takeaways and encountered challenges are:

Learnings

➢ *Pattern Discovery:* Clustering helped uncover hidden patterns and structures within taxi trip data, revealing distinct customer segments and spatial clusters of taxi usage.

➢ *Geospatial Visualization:* Visualizing clusters on maps provided a geographic understanding of taxi pickup patterns, highlighting areas of high and low demand.

➢ *Segment Characteristics:* Analyzing cluster characteristics like average trip distance and passenger count deepened our understanding of customer behaviors and preferences.

➢ *Operational Insights:* Clustering insights offered actionable intelligence for route optimization, resource allocation, and service enhancements tailored to specific customer segments.

Challenges

➢ *Data Quality and Preprocessing:* Considering the large size, cleaning and preprocessing raw taxi trip data, handling missing values, and ensuring data integrity posed initial challenges.

➢ *Optimal Cluster Selection:* Determining the optimal number of clusters using techniques like the elbow method required iterative experimentation and interpretation.

➢ *Interpretability:* Interpreting results and translating them into actionable insights for stakeholders required careful analysis and domain knowledge.

➢ *Scalability:* Scaling algorithms to handle large datasets efficiently was a technical challenge that required optimization.

FUTURE SCOPE

Our exploration of NYC taxi trip data through clustering opens avenues for future enhancements and applications in urban transportation analysis:

➢ *Real-time Predictive Models:* Implementing real-time predictive models using time series analysis and clustering for dynamic taxi service optimization.

➢ *Integration of External Factors:* Incorporating external factors like weather conditions, events, or traffic data to enrich cluster analysis and enhance predictive capabilities.

➢ *Advanced Spatial Analysis:* Exploring advanced spatial analytics techniques to uncover more profound insights into regional variations and hotspots of taxi

demand.
➢ *Collaborative Decision Support Systems:* Integrating clustering insights into decision support systems for policymakers, urban planners, and transportation authorities.

Our analysis of NYC taxi trip data using clustering techniques has provided actionable insights and strategic recommendations for optimizing taxi services and enhancing urban mobility. By leveraging insights, we have identified distinct customer segments, seasonality, spatial clusters of taxi usage, and critical patterns influencing trip behaviors. These findings can inform evidence-based decision-making, resource allocation, and service improvements within NYC's taxi industry.

The application of advanced analytics techniques underscores the value of data-driven approaches in understanding and optimizing urban transportation systems, paving the way for future innovations and advancements in this dynamic field. Our journey highlights the transformative potential of data analytics in shaping the future of urban mobility, ensuring efficient, sustainable, and customer-centric transportation services for all.

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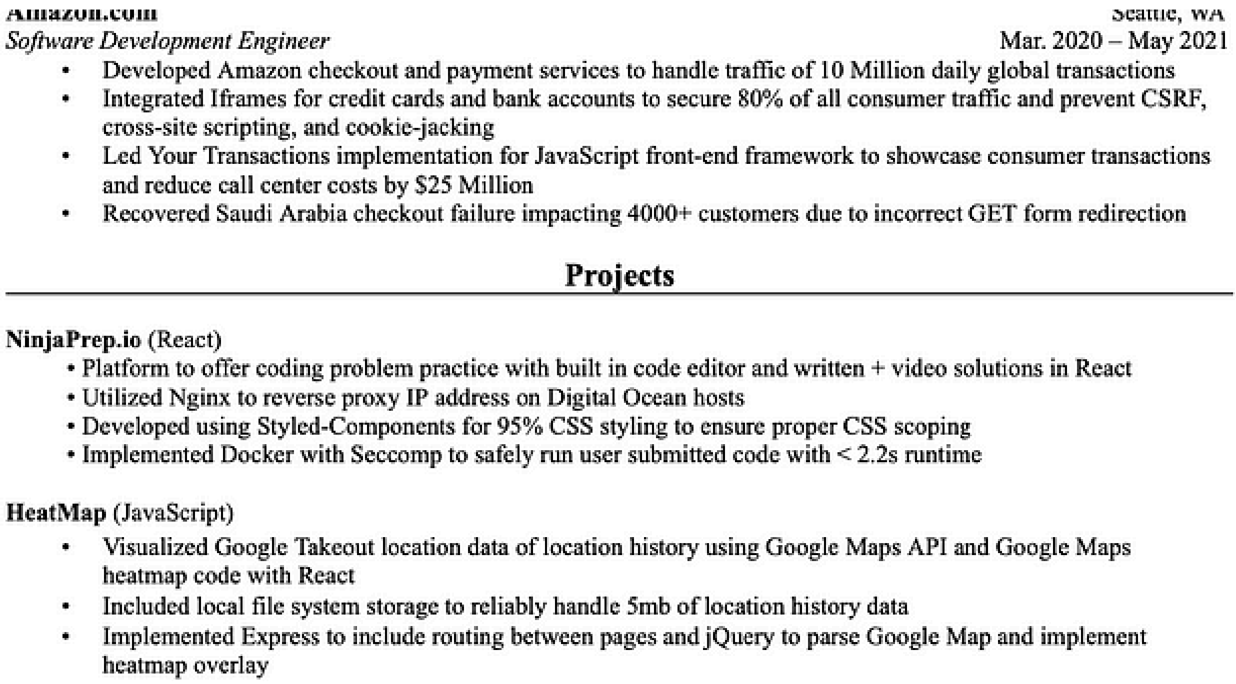

Amazon.com Seattle, WA
Software Development Engineer Mar. 2020 – May 2021

- Developed Amazon checkout and payment services to handle traffic of 10 Million daily global transactions
- Integrated Iframes for credit cards and bank accounts to secure 80% of all consumer traffic and prevent CSRF, cross-site scripting, and cookie-jacking
- Led Your Transactions implementation for JavaScript front-end framework to showcase consumer transactions and reduce call center costs by $25 Million
- Recovered Saudi Arabia checkout failure impacting 4000+ customers due to incorrect GET form redirection

Projects

NinjaPrep.io (React)
- Platform to offer coding problem practice with built in code editor and written + video solutions in React
- Utilized Nginx to reverse proxy IP address on Digital Ocean hosts
- Developed using Styled-Components for 95% CSS styling to ensure proper CSS scoping
- Implemented Docker with Seccomp to safely run user submitted code with < 2.2s runtime

HeatMap (JavaScript)
- Visualized Google Takeout location data of location history using Google Maps API and Google Maps heatmap code with React
- Included local file system storage to reliably handle 5mb of location history data
- Implemented Express to include routing between pages and jQuery to parse Google Map and implement heatmap overlay

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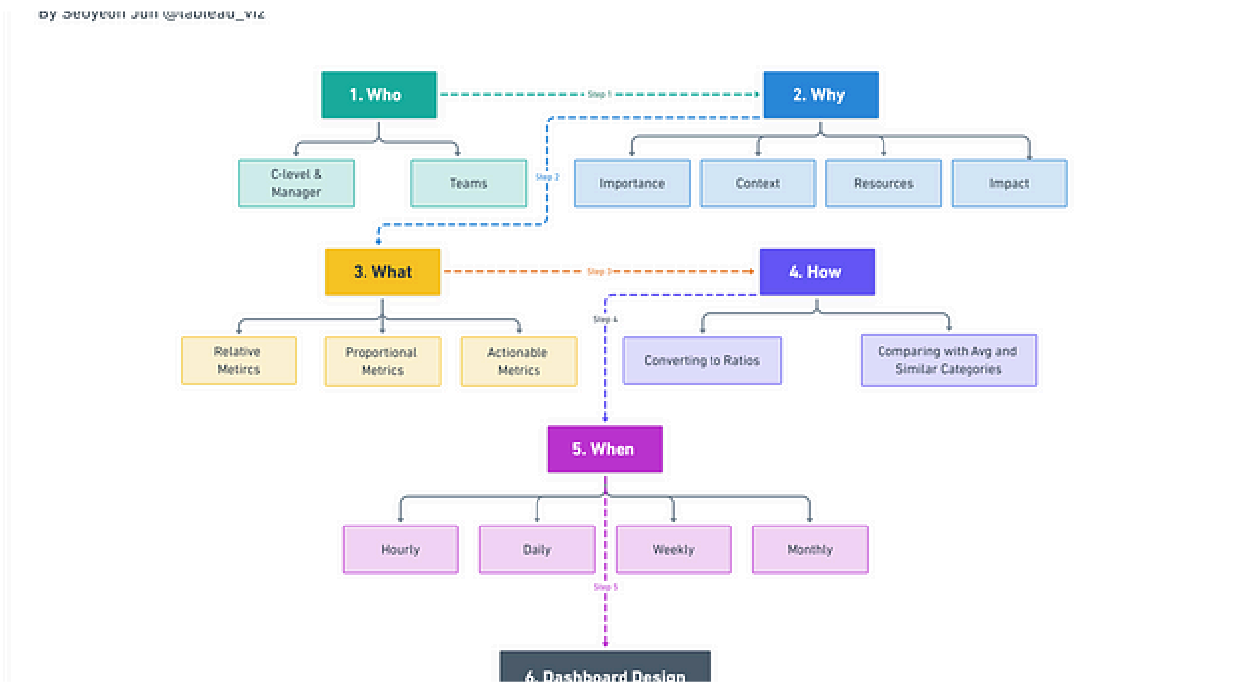

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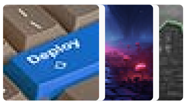

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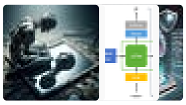

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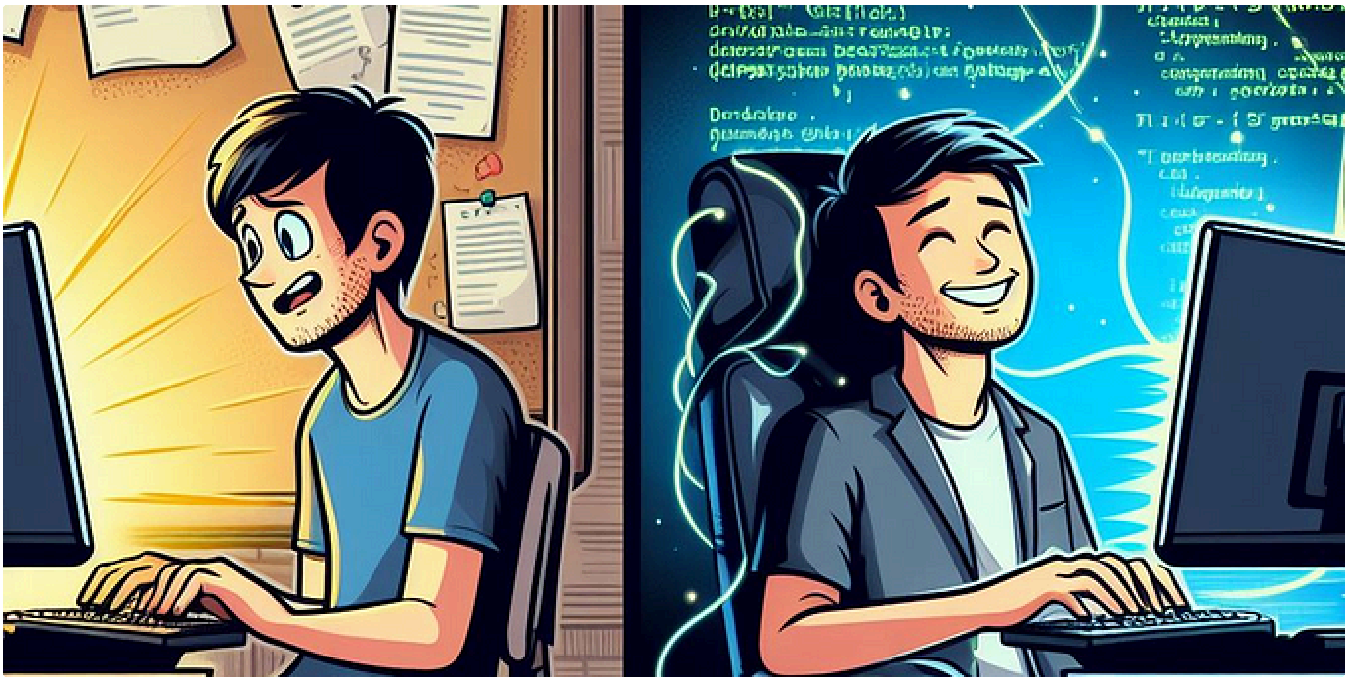

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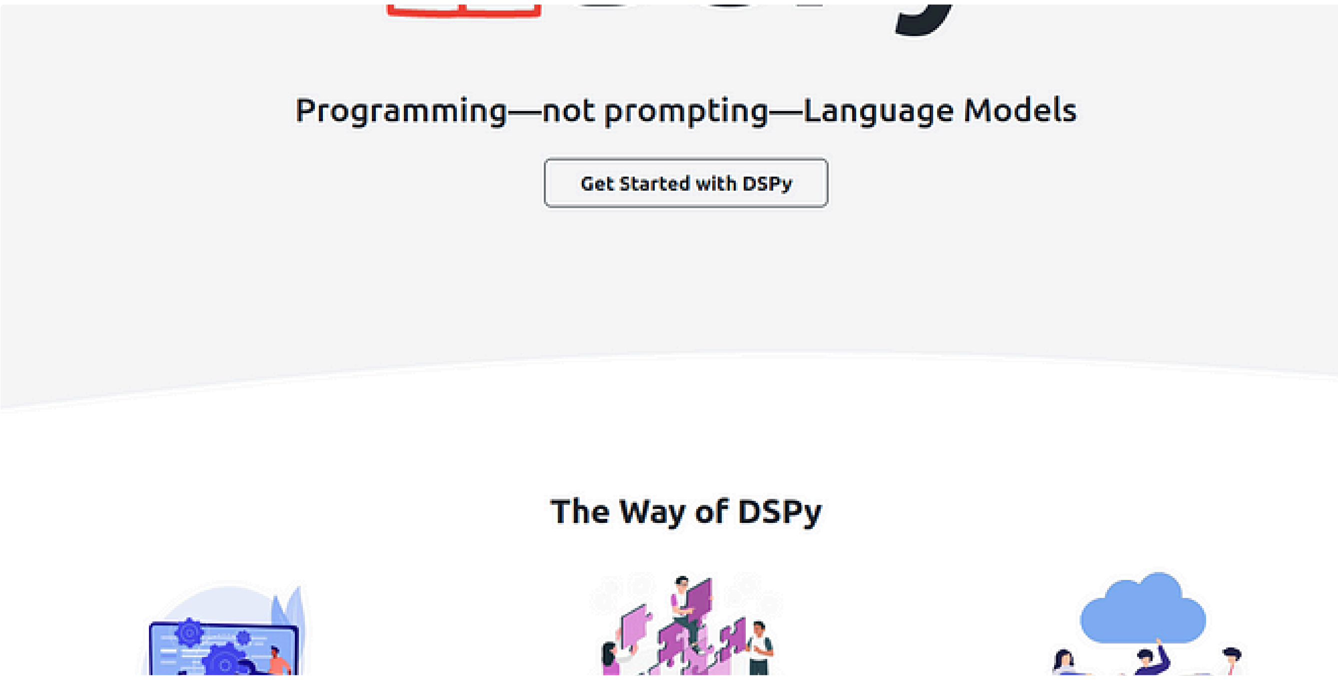

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